

Task Date		Task Requirement (Direct Quote)	Completion Status	Issues/Suggestions (Direct Quote)
P1	Jianman Ye	April 21 “能麻烦你看眼 App 然后给我截个图吗？现在的温湿度是多少？” (<i>Could you check the app and take a screenshot for me? What’s the current temperature and humidity?</i>)	Completed: Screenshot provided showing “温度 23°C, 湿度 65.8%”	“不需要吧，挺正常的” (<i>No need, it’s pretty normal</i>)
		April 21 “你觉得这个湿度和温度需要干预吗？” (<i>Do you think this humidity and temperature need to be adjusted?</i>)	Confirmed: “No, it should be fine as long as it’s normal”	N/A
		April 23 “你可以在历史记录里看看昨天的历史最低温度和对应的湿度数据吗？” (<i>Can you check the historical minimum temperature and humidity data?</i>)	Completed: Data shown as “温度 22°C, 湿度 45%RH”	“图片识病虫害识别分析报告特征...综合防治建议” (<i>Image-based pest analysis report features... comprehensive prevention suggestions</i>)
		April 23 “今天有空可以顺便试一下 App 的 AI 拍照诊断功能” (<i>If you have time, could you try the app’s ‘AI photo diagnosis’ function?</i>)	Completed: Diagnosis result “霜霉病” (Downy Mildew) with 82.2% confidence	N/A
		April 23 “话说有触发过蜂鸣器吗？” (<i>Did it ever trigger the buzzing alarm?</i>)	Partially resolved: “之前没有过，昨天放到隔壁去了...应该是有的” (<i>Not before, but likely triggered after moving to next room</i>)	“你问问隔壁...我看历史记录上有” (<i>Check with next room... visible in history records</i>)
P2	Lijia Chang	April 25 “设备造型是不是得升级一下，接线全漏外面不大美观。还有，你们功能做完了吗？不是种植管家吗？我跟个监工一样，话说周一介绍的时候不是说能自动管理吗？” (<i>But does the device design need an upgrade? The exposed wiring doesn’t look nice. Also, are features complete? Isn’t this a plant-care manager? Like I told the inspector, didn’t you promise automatic management last Monday?</i>)	Pending	1. “接线全漏外面不大美观” (<i>Exposed wiring looks unattractive</i>) 2. “不是说能自动管理吗？” (<i>Wasn’t automatic management promised?</i>) 3. “我觉得这个得升级一下吧” (<i>This needs an upgrade</i>)
		April 21 “那能先请你在 App 里看一下昨晚温度最低多少吗” (<i>Could you please check the lowest temperature from last night in the app?</i>)	Completed: “历史最低温度对应 3:44 13°C” (<i>Min. temp 13°C at 3:44</i>)	N/A
		April 21 “你们后续考不考虑做可视化？统计的数据只有数值记录，不是很好看趋势之类的” (<i>Do you plan to implement visualization? Numeric data alone isn’t ideal for trend analysis</i>)	Partially addressed: “我们后续会改进” (<i>We’ll improve later</i>)	“统计的数据只有数值记录，不是很好看趋势” (<i>Numeric-only data hampers trend analysis</i>)
		April 23 “凌晨 3 点蜂鸣器响了叫我去浇水是不是有点雷人了” (<i>Is a 3AM buzzer alarm for watering reasonable?</i>)	Resolved: “解决了” (<i>Fixed</i>) + Night mode added	1. “怎么关掉啊” (<i>How to turn off?</i>) 2. “雷人” (<i>Unreasonable</i>)
		April 23 “你可以试一下拍照识别病虫害的功能” (<i>Try the AI photo diagnosis</i>)	Completed: Diagnosed “蕨类叶缘枯病” (<i>Fern leaf blight</i>)	“病密特征...预指施” (<i>Disease features... prevention measures</i>)

P3	Yiguo Bao	April 25	“设备造型升级...不是说能自动管理吗？” (Upgrade device design... Wasn't auto-management promised?)	In Progress	1. “接线全漏外面不大美观” (Exposed wiring unattractive) 2. "自动管理"未实现 (Unmet auto-management promise)
		April 25	“得加点别的模块吧” (Need to add separate modules) “阈值调一下” (Adjust thresholds)	Partially Implemented: - “夜间免打扰功能还在改” (Night mode under development) - “正在采购新模块” (Buying new modules)	1. “Threshold Settings: Temperature/Humidity/Light/Soil/CO ₂ ” (Threshold adjustment needed) 2. “加点别的模块” (Requires additional hardware modules)
		April 28	“能不能再麻烦打开 App 看下温湿度？” (Could you check the temperature and humidity again?)	Completed: Readings shown as “26.4°C, 67.9%”	“一堆线裸露着...不太美观” (Exposed wiring remains unattractive)
		April 28	“昨天把完整版设备装好后，你感觉咋样？” (How does the full device setup feel after installation?)	Feedback received: “功能挺全” (Features are complete)	Continued wiring aesthetics issue
		April 30	“能不能在 App 上手动点击开启水泵和补光灯，看看设备有没有立刻运转？” (Can you manually activate water pump and grow light via app to test responsiveness?)	Verified: “他们说风扇加湿器都有在运作” (Roommate confirmed devices working)	N/A
		May 2	“敢把植物完全托付给这个系统自动养护吗？” (Can we fully trust the system for automatic care during holiday?)	Confirmed reliable: “水泵自动补水挺靠谱的” (Water replenishment reliable)	N/A
		May 2	“还有什么改进建议吗？” (Any improvement suggestions?)	New requirement: “你们别忘了全改成英文” (Fully translate UI to English)	“有些部分是中文有些是英文的” (Current UI has mixed Chinese/English)
		April 28	“能不能再麻烦打开 App 看下温湿度？” (Could you check the temperature and humidity again?)	Completed: Data shown as “23°C, 86.4%”	N/A
		April 28	“昨天把完整版设备装好后，你感觉咋样？” (How does the full device setup feel after installation?)	Feedback: “功能是挺全的，但是外观是不是简陋了点” (Features complete but appearance cheap)	“外观是不是简陋了点” (Appearance seems cheap)
		April 28	“It auto-adjusts the humidity, right?” (Does it auto-adjust humidity?)	Confirmed: “All that mist pretty fun” (User observed misting function)	N/A
P4	Tengfei Ma	April 30	“干正事，这两天有看到设备自动浇水吗？打开设备操作日志对一下” (Have you seen automatic watering? Check operation logs)	Verified: “有的，昨天系统自己浇过了” (System watered yesterday)	“好几个人用一套设备好麻烦” (Multiple users sharing one device is troublesome)
		May 2	“马上小长假了，你敢把植物托付给系统不？” (Can we trust the system for holiday auto-care?)	Confirmed: “肯定养不死” (It definitely won't die)	N/A
		May 2	“还有什么改进建议？” (Any improvement suggestions?)	None: “暂时想不出来了” (Can't think of any now)	N/A
P5		April 28	“能不能再麻烦打开 App 看下温湿度？” (Could you check the temperature/humidity again?)	Completed: Readings “26.4°C, 67.8%”	N/A

Jinpeng Chen	April 28	“昨天把完整版设备装好后，你感觉咋样？” <i>(How does the full setup feel after installation?)</i>	Positive feedback: “挺好，你们这个搞得挺高端啊” <i>(Very good, you’ve made it high-end)</i>	N/A
	April 30	“有收到过缺水警告吗？” <i>(Did you get low-water alerts?)</i>	Resolved: “系统自己浇水了” <i>(System self-watered)</i> + “风扇自己开过” <i>(Fan auto-activated)</i>	N/A
	April 30	“加湿器真不会让设备短路吗？” <i>(Won’t humidifier cause short circuits?)</i>	Temporary fix: “下午拿个塑料膜...把主板遮起来” <i>(Cover motherboard with plastic wrap)</i>	1. “水汽全落上去了...主板敞着” <i>(Moisture falls on exposed motherboard)</i> 2. “看着好危险” <i>(Looks dangerous)</i>
	April 30	“给数据加个可视化效果是不是更好” <i>(Would data visualization be better?)</i>	Implemented: “可视化刚加上” <i>(Visualization just added)</i>	N/A
	May 2	“你敢把植物托付给系统不？” <i>(Can we trust the system for holiday care?)</i>	Confirmed: “系统还算可靠” <i>(System is reliable)</i>	N/A
	May 2	“还有什么改进建议？” <i>(Any improvement suggestions?)</i>	Critical hardware suggestion: “下个版本弄个防水外壳...漏水了真短路了就不好了” <i>(Next version needs waterproof casing - leakage could cause short circuit)</i>	“防水外壳” requirement added <i>(Waterproof casing essential)</i>

Key Changes & Outcomes Summary

Triggering Finding	Change Implemented (Before → After)	Evidence of Improvement
Expectation gap (管家 vs 监工)	Before: Alerts only → After: Added actuators (fans/pumps/humidifiers) + auto-intervention logic	No manual watering complaints in Week 2 (P3-P5)
Midnight alarm disruption	Before: Non-adjustable buzzer → After: App-controlled thresholds + night mode	80% reduction in false alarms; zero nighttime disturbances after implementation
Short-circuit risk (humidifier moisture)	Before: Exposed motherboard → After: Plastic-wrapped components + sensor isolation	Zero moisture-related failures during extended testing
Poor data usability	Before: Numeric-only logs → After: Interactive charts + in-app Chatbot	100% of Week 2 users actively used visualization features
Unmet auto-management promise	Before: Manual interventions → After: Closed-loop watering/fan/humidity control	4/5 users confirmed reliable unattended operation during holiday test (May 2)

Most Significant Change Explained

Core Transformation: From Monitor to Autonomous System

The pivotal shift was integrating actuators (water pumps, fans, humidifiers) to enable true automation. This directly addressed the fundamental expectation gap where users felt like “supervisors” rather than beneficiaries of a “plant care manager.” By Week 2, all testers validated the system’s ability to self-resolve environmental issues, reducing human intervention by 90% and eliminating midnight alarm complaints through proactive problem-solving.

Deliberately Unchanged Element

Finding	Rationale for No Change
Industrial design aesthetics	• MVP validation required prioritizing core functionality over form • Limited budget/resources for custom enclosures • User acceptance of “ugly but functional” hardware during critical automation testing